Ecology and the environment

Mark scheme and worked solutions for 25 questions: 8 multiple choice, 17 structured.

1 · Part 1

Paper 1 & 2 | The organism in the environment | [1]

Answer: B

Explanation

A community is all the populations of different species living together in the same area. A population is all the organisms of one species in an area, a habitat is the place where an organism lives, and an ecosystem is a community together with its physical (abiotic) environment, so A, C and D are wrong.

2 · Part 1

Paper 2 only | The organism in the environment | [1]

Answer: D

Explanation

Biodiversity is the variety of living organisms in an area, which includes how many different species are present. A describes the size of a single population, B describes biomass, and C describes a habitat.

3 · Part 1

Paper 1 & 2 | Feeding relationships | [1]

Answer: A

Worked solution

Count along the chain: producer (phytoplankton), primary consumer (zooplankton), secondary consumer (herring), tertiary consumer (seal).

4 · Part 1

Paper 1 & 2 | Feeding relationships | [1]

Answer: C

Worked solution

Seaweed to limpets: 10% of 15 000 kJ = 1 500 kJ. Limpets to crabs: 10% of 1 500 kJ = 150 kJ. Check: 15 000 × 0.1 × 0.1 = 150 kJ.

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5 · Part 1 Paper 2 only Cycles within ecosystems [1]

Answer: C

Explanation

Nitrifying bacteria convert ammonium ions into nitrates. Nitrogen fixing bacteria convert nitrogen gas into nitrogen compounds, denitrifying bacteria convert nitrates into nitrogen gas, and decomposers break down proteins in dead organisms and waste to release ammonium ions.

6 · Part 1 Paper 1 & 2 Human influences on the environment [1]

Answer: A

Explanation

Water vapour is one of the greenhouse gases (with carbon dioxide, nitrous oxide, methane and CFCs). Oxygen, nitrogen and hydrogen are not greenhouse gases.

7 · Part 1 Paper 1 & 2 Human influences on the environment [1]

Answer: D

Explanation

Carbon monoxide combines with haemoglobin in red blood cells, so less haemoglobin is available to carry oxygen and the tissues receive less oxygen. Carbon monoxide does not lower blood pH (A), does not react with glucose (B) and does not stop carbon dioxide being carried in the plasma (C).

8 · Part 1 Paper 2 only Human influences on the environment [1]

Answer: B

Explanation

Tree roots hold soil in place and the tree canopy shelters the ground from heavy rain; when the trees are removed, rain washes the soil away. Ash does not make soil particles smaller (A), the drying of soil for decomposers does not explain erosion (C), and soil does not dissolve because of carbon dioxide in the air (D).

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9 · Part 1

Paper 1 & 2 | The organism in the environment | [5]

Mark scheme

(a) **[1]**

- the (35) common frogs (1)

ALLOW the water snails

ALLOW the dragonfly larvae

REJECT water weed (several species)

(b) **[2]**

- a community of organisms OR all the living organisms (populations) in an area (1)
- together with their physical (abiotic) environment and the interactions between them (1)

ALLOW non-living parts of the area such as water, mud and stones for the physical environment

(c) **[2]**

- less light reaches the water weed so there is less photosynthesis (1)
- so less growth OR less reproduction, so the population of water weed decreases (1)

ALLOW some water weed plants die

Worked solution

(a) A population is all the organisms of one species living in an area. The 35 common frogs are one species, so they form a population. The water weed is several different species, so it is not one population.

(b) An ecosystem is the community of living organisms in an area together with their physical environment, and the way they interact. In this pond the community (frogs, snails, water weed, dragonfly larvae) and the water, mud and stones together make up the ecosystem.

(c) Light intensity is an abiotic factor. Shade from the tree means the water weed receives less light, so it photosynthesises less and makes less glucose for growth. The plants grow and reproduce less, and some may die, so the population size of the water weed falls.

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10 · Part 1

Paper 1 & 2 The organism in the environment [8]

Mark scheme

(a) **[2]**

- lay out two tape measures at right angles along the edges of the area to make a grid (1)
- use random numbers (from a random number generator or table) as coordinates to place each quadrat (1)

IGNORE throwing the quadrat over the shoulder for the first mark unless coordinates are used

ALLOW pick numbered squares of a grid at random for the second mark

(b) **[1]**

- 7.0 (daisy plants per quadrat) (1)

ALLOW 7

(c) **[3]**

- area of one quadrat = 0.5 × 0.5 = 0.25 m^2 (1)
- 600 ÷ 0.25 = 2 400 (1)
- 2 400 × 7.0 = 16 800 (daisy plants) (1)

ECF from (b) for the final mark

ALLOW 16 800 with no working for 3 marks

(d) **[2]**

- the tall grass shades the daisy plants (in area Y) (1)
- so they get less light for photosynthesis OR they cannot compete with the grass for light (1)

ALLOW competition with grass for water or mineral ions for the second mark

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Worked solution

(a) Random placing avoids choosing places where there seem to be more or fewer daisies. Make a grid with two tapes at right angles, then use pairs of random numbers as coordinates for each quadrat.

(b) Total in area X = 6 + 9 + 4 + 11 + 7 + 8 + 5 + 10 + 6 + 4 = 70. Mean = 70 ÷ 10 = 7.0 daisy plants per quadrat.

(c) Each quadrat is 0.5 m × 0.5 m = 0.25 m^2. The number of quadrats that would fit into area X is 600 ÷ 0.25 = 2 400. Estimated population = 7.0 × 2 400 = 16 800 daisy plants. Check: 7.0 daisies per 0.25 m^2 is 28 per m^2, and 28 × 600 = 16 800.

(d) In area Y the grass is tall. Daisies are short plants, so the grass shades them and they receive less light. They photosynthesise less, grow less and are outcompeted, so fewer daisy plants survive there.

11 · Part 1

Paper 1 & 2 The organism in the environment [6]

Mark scheme

(a) **[3]**

- the number of aphids increases from week 0 to week 6 (1)
- reaches a maximum of 210 per plant at week 6 (1)
- then decreases to 40 per plant by week 12 (1)

ALLOW the decrease slows down between week 10 and week 12 for the third mark

At least one data value must be quoted for full marks.

(b) **[3]**

- the number of ladybirds increases from 12 to a maximum of 44 at week 8 and then falls to 20 at week 10 (1)
- increase because there are more aphids, so more food, so more ladybirds survive and reproduce (1)
- decrease because the number of aphids has fallen, so there is less food OR more competition for food between ladybirds, so ladybirds die OR reproduce less (1)

ALLOW ladybirds leave the greenhouse to find food

Worked solution

(a) The mean number of aphids rises from 20 per plant at week 0 to a peak of 210 per plant at week 6. It then falls to 120 at week 8 and 50 at week 10, and only falls slightly to 40 by week 12.

(b) Aphids are the ladybirds' food (a biotic factor). As the aphid population grows, the ladybirds have plenty of food, so more survive and breed: their number rises from 12 at week 4 to 44 at week 8. The ladybirds eat many aphids, so the aphid number falls after week 6. With fewer aphids, the ladybirds compete for less food, so many die or breed less, and their number falls to 20 at week 10. The ladybird peak comes after the aphid peak because the predators need time to breed.

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12 · Part 1

Paper 2 only | The organism in the environment | [8]

Mark scheme

(a) **[2]**

Any two of:

- the size of the quadrat (1)
- the same person estimating percentage cover OR the same method of estimating percentage cover (1)
- light measured at the same time of day OR in the same weather conditions (1)
- light meter held at the same height OR position in each quadrat (1)

REJECT light intensity (this is the factor being measured)

(b) **[2]**

- as light intensity increases, the percentage cover of grass increases (1)
- data quoted, e.g. 5% at 1 000 lux and 90% at 20 000 lux (1)

ALLOW positive correlation for the first mark

(c) **[2]**

- light intensity increases further from the wood (because there is less shade from trees) (1)
- so grass can photosynthesise more, so it grows more OR produces more glucose for growth (1)

ALLOW grass outcompetes moss in high light for the second mark if photosynthesis is mentioned

(d) **[2]**

- the data support a link: moss cover is highest where light intensity is lowest OR moss cover decreases as light intensity increases (1)
- but other factors also change with distance, e.g. soil moisture OR humidity OR temperature OR competition from grass, so light may not be the only factor (1)

ALLOW only one line of quadrats was used, so the conclusion is not reliable, as an alternative second mark

Worked solution

(a) To make the quadrats a fair comparison, everything except the position should stay the same: the quadrat size, the way percentage cover is estimated, and how and when light is measured (same time of day, same height of the light meter).

(b) Grass cover rises steadily as light intensity rises, from 5% at 1 000 lux to 90% at 20 000 lux.

(c) Near the wood the trees shade the ground. Further into the field there is more light, so grass photosynthesises faster, grows more and covers more of the ground.

(d) The data do show that moss is most common where light is low. However, under the trees other abiotic factors such as soil moisture and humidity are also different, and moss in the field also has to compete with grass (a biotic factor). These factors change along the same line, so the data cannot show that light is the only factor. The conclusion is only partly supported.

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13 · Part 1

Paper 2 only | The organism in the environment | [7]

Mark scheme

(a) **[2]**

- meadow P has a higher biodiversity than meadow Q (1)
- P has 6 species and Q has 3 species (1)

> **ALLOW** in P the plants are spread more evenly between species, while in Q most plants are ryegrass, as an alternative second mark

(b) **[2]**

- (150 ÷ 160) × 100 (1)
- 94 (%) (1)

> **ALLOW** 93.75 with no rounding for 1 mark only

(c) **[2]**

- fertiliser allows ryegrass to grow faster OR taller (1)
- so it outcompetes the other species for light OR space, so the other species die out (1)

(d) **[1]**

- so that the results from the two meadows can be compared fairly OR it is a fair test (1)

> **ALLOW** so that the same area is sampled in each meadow

Worked solution

(a) Biodiversity is the variety of species in an area. Meadow P has 6 species and the plants are shared fairly evenly between them. Meadow Q has only 3 species and most of its plants are ryegrass, so meadow Q has a lower biodiversity.

(b) Percentage = (150 ÷ 160) × 100 = 93.75%, which is 94% to the nearest whole number. Check: the other two species make up 10 ÷ 160 × 100 = 6.25%, and 100 − 6.25 = 93.75%.

(c) The fertiliser provides extra mineral ions. Ryegrass grows quickly when it has them, so it becomes tall and dense, shading the smaller plants and taking more space. The other species cannot compete and die out, so fewer species remain.

(d) If one meadow had more or larger quadrats, more plants and probably more species would be found there just because a bigger area was sampled. Using the same number and size of quadrats means the comparison is fair.

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14 · Part 1

Paper 1 & 2 | Feeding relationships | [5]

Mark scheme

(a) **[1]**

- grass (1)

(b) **[1]**

- grass → grasshopper → frog → grass snake (1)

ALLOW plural names, e.g. grasshoppers, frogs, grass snakes

REJECT arrows pointing the wrong way

(c) **[1]**

- tertiary consumer (1)

ECF from a four-organism chain in (b)

(d) **[2]**

- the number of grasshoppers increases (1)
- because fewer grasshoppers are eaten (by frogs) (1)

ALLOW later the number may fall again because there is less grass OR more competition for food OR more spiders eat them

Worked solution

(a) The producer is the organism that makes its own food by photosynthesis: grass.

(b) Follow the "is eaten by" links from grass: grass → grasshopper → frog → grass snake. The arrows show the direction in which energy and substances pass.

(c) Grass is the producer, grasshoppers are primary consumers, frogs are secondary consumers and grass snakes are tertiary consumers.

(d) Frogs are predators of grasshoppers. If most frogs die, fewer grasshoppers are eaten, so the grasshopper population increases. Later it may fall again because the grasshoppers run short of grass or because spiders, with more prey available, eat more of them.

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15 · Part 1 Paper 1 & 2 Feeding relationships [6]

Mark scheme

(a) **[2]**

- pyramid of numbers is not a pyramid shape: the bar for apple trees is smaller than the bar for caterpillars (because a few large trees support many caterpillars) (1)
- pyramid of biomass is a pyramid shape: each bar is smaller than the one below it (1)

(b) **[2]**

- (70 ÷ 7 000) × 100 (1)
- 1.0 (%) (1)

ALLOW 1 (%)

(c) **[2]**

- blue tits eat caterpillars and digest them (into small soluble molecules) (1)
- the products are absorbed and used to build new cells OR tissues OR for growth in the blue tits (1)

ALLOW examples such as amino acids used to make proteins

Worked solution

(a) Each apple tree is very large and feeds thousands of caterpillars, so in a pyramid of numbers the bottom bar (10 trees) is much narrower than the middle bar (25 000 caterpillars). In a pyramid of biomass, the bars show mass: 2 000 000 g, then 7 000 g, then 70 g, so each bar is smaller than the one below and the shape is a true pyramid.

(b) Percentage = (70 ÷ 7 000) × 100 = 1.0%. Check: 1% of 7 000 g is 70 g.

(c) Substances pass along a food chain by feeding. The blue tits eat the caterpillars, digest the large molecules such as proteins into small soluble molecules, absorb them into the blood and use them to build their own cells and tissues.

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16 · Part 1 Paper 1 & 2 Feeding relationships [8]

Mark scheme

(a) **[2]**

- (2 160 ÷ 24 000) × 100 (1)
- 9.0 (%) (1)

ALLOW 9 (%)

(b) **[4]**

Any four of:

- not all the grass is eaten, e.g. roots are left (1)
- some of the grass eaten is not digested OR is lost in faeces (1)
- energy is released in respiration by the sheep (1)
- this energy is lost (to the surroundings) as heat (1)
- energy is used for movement OR keeping the body warm (1)
- some energy is lost in urine (1)

(c) **[2]**

- there is one less trophic level in the food chain wheat → people (1)
- at each transfer most of the energy (about 90%) is lost, so feeding wheat to sheep first loses much of the energy before it reaches people (1)

ALLOW people get about ten times more energy from eating the wheat directly

Worked solution

(a) Percentage = (2 160 ÷ 24 000) × 100 = 9.0%. Check: 9% of 24 000 kJ = 0.09 × 24 000 = 2 160 kJ.

(b) Most of the energy in the grass never becomes sheep biomass. The sheep do not eat all the grass, for example the roots. Some of the grass they eat cannot be digested and passes out in the faeces. Much of the energy that is absorbed is released in respiration and used for movement and keeping warm, and this energy is lost to the surroundings as heat. A little is lost in urine. Only a small amount is left to make new biomass.

(c) In the chain wheat → sheep → people, about 90% of the energy in the wheat is lost at the sheep stage. In the chain wheat → people, that step is missing, so far more of the energy in the wheat reaches people.

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17 · Part 1

Paper 1 & 2 | Cycles within ecosystems | [7]

Mark scheme

(a) **[1]**

- photosynthesis (1)

(b) **[3]**

- decomposers OR bacteria and fungi feed on the dead leaf (1)
- they break down (decompose) the carbon compounds in the leaf (1)
- the decomposers respire and release carbon dioxide into the atmosphere (1)

(c) **[2]**

- little oxygen, so few decomposers OR decomposers cannot respire aerobically (1)
- so the dead plants are not decomposed (or are decomposed only very slowly), so carbon is not released as carbon dioxide (1)

ALLOW low oxygen slows down decomposition

(d) **[1]**

- combustion (1)

ALLOW burning

Worked solution

(a) Plants take carbon dioxide from the air and use it to make glucose in photosynthesis.

(b) When a leaf dies, decomposers such as bacteria and fungi feed on it. They break down the carbon compounds (carbohydrates, proteins and lipids) in the leaf and use them in respiration. Respiration releases carbon dioxide, which returns the carbon to the atmosphere.

(c) Decomposers need oxygen to respire and decompose dead material quickly. In waterlogged peat there is very little oxygen, so decomposition is very slow. The carbon stays locked up in the peat instead of being released as carbon dioxide.

(d) Burning peat is combustion. The carbon compounds in the peat react with oxygen and release carbon dioxide into the atmosphere.

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18 · Part 1 Paper 2 only Cycles within ecosystems [7]

Mark scheme

(a) **[2]**

- convert nitrogen gas from the air (1)
- into nitrogen compounds, e.g. ammonium ions OR nitrates, that plants can use (1)

ALLOW to make proteins OR amino acids for the second mark if nitrogen compounds are stated

(b) **[2]**

- decomposers break down the proteins (or amino acids) in the dead clover (1)
- releasing ammonium ions into the soil (1)

ALLOW ammonia for ammonium ions

(c) **[3]**

- denitrifying bacteria are more active when there is little oxygen (1)
- they convert nitrates into nitrogen gas (which is lost to the air) (1)
- nitrifying bacteria need oxygen, so fewer ammonium ions are converted into nitrates (1)

Worked solution

(a) Plants cannot use nitrogen gas from the air. Nitrogen fixing bacteria, such as those in the clover root swellings, convert nitrogen gas into nitrogen compounds such as ammonium ions, which the plant uses to make amino acids and proteins.

(b) The dead clover contains proteins. Decomposers (bacteria and fungi) break down these proteins and release ammonium ions into the soil. These can then be turned into nitrates.

(c) In waterlogged soil there is very little oxygen. Denitrifying bacteria thrive in these conditions and convert nitrates into nitrogen gas, which escapes to the air. Nitrifying bacteria need oxygen, so they make fewer nitrates from ammonium ions. Both effects mean less nitrate is available to the crop.

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19 · Part 1

Paper 1 & 2 | Human influences on the environment | [6]

Mark scheme

(a) **[2]**

- as the concentration of sulfur dioxide decreases, the number of lichen species increases (1)
- data quoted, e.g. 1 species at 95 µg per m^3 and 14 species at 5 µg per m^3 (1)

ALLOW negative correlation for the first mark

(b) **[3]**

- sulfur dioxide dissolves in water in the clouds OR rain (1)
- forming acid rain (1)
- acid rain lowers the pH of the lake, which kills fish OR other organisms in the lake (1)

ALLOW acid rain damages the leaves of plants around the lake

ALLOW sulfur dioxide is carried by the wind to the lake area

(c) **[1]**

- other factors also change with distance from the power station, e.g. other pollutants OR the number of trees OR the type of bark (1)

ALLOW only one set of data OR no repeats

Worked solution

(a) Near the power station the sulfur dioxide concentration is highest (95 µg per m^3) and only 1 lichen species is found. Far away it is lowest (5 µg per m^3) and 14 species are found. The lower the sulfur dioxide concentration, the more lichen species there are.

(b) Wind carries sulfur dioxide away from the power station. It dissolves in water in clouds and falls as acid rain. When acid rain falls on the lake and its surroundings, the water becomes more acidic (lower pH), and fish and other aquatic organisms are harmed or killed.

(c) The data show a correlation, not a cause. Other things such as other pollutants from the power station may also change with distance and could affect the lichens.

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20 · Part 1 Paper 1 & 2 Human influences on the environment [6]

Mark scheme

(a) **[4]**

- raising cattle: methane (1)
- burning petrol: carbon dioxide (1)
- adding nitrogen-containing fertiliser: nitrous oxide (1)
- leaks from old refrigerators: CFCs (1)

ALLOW chlorofluorocarbons for CFCs

ALLOW water vapour for burning petrol

REJECT carbon monoxide for burning petrol

(b) **[2]**

- more people need more food, so more cattle are farmed OR more rice is grown (in paddy fields) (1)
- cattle OR rice fields OR landfill sites release methane (1)

ALLOW more waste is put in landfill sites, where it decomposes and releases methane

Worked solution

(a) Cattle produce methane in their digestive systems. Burning petrol releases carbon dioxide. Fertilisers containing nitrogen lead to the release of nitrous oxide from the soil. Old refrigerators leak CFCs.

(b) As the population grows, more food is needed. More cattle are farmed and more rice is grown in flooded fields, and both release methane. More rubbish also goes to landfill, where its decomposition releases methane.

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21 · Part 1 Paper 1 & 2 Human influences on the environment [7]

Mark scheme

(a) **[4]**

- energy from the Sun reaches the Earth's surface and warms it (1)
- the Earth's surface gives out heat (infrared radiation) (1)
- greenhouse gases absorb this heat OR stop it escaping into space OR re-radiate it back to the Earth (1)
- more greenhouse gases means more heat is kept in the atmosphere, causing the enhanced greenhouse effect (1)
- which increases the mean temperature of the Earth (global warming) (1)

Maximum 4 marks.

(b) **[1]**

- 250 m (1)

(c) **[2]**

- global warming has made the lower slopes warmer (1)
- species M can only survive in cold conditions, so it now survives only at higher altitudes OR other species that grow better in warmer conditions outcompete it at lower altitudes (1)

Worked solution

(a) Energy from the Sun passes through the atmosphere and warms the Earth's surface. The warm surface gives out heat as infrared radiation. Greenhouse gases absorb much of this heat and send some of it back towards the Earth, which keeps the Earth warm. If the concentration of greenhouse gases increases, more heat is trapped. This is the enhanced greenhouse effect, and it raises the mean temperature of the Earth, which is global warming.

(b) Increase = 2 050 − 1 800 = 250 m.

(c) Higher altitudes are colder. As global warming raises temperatures, the lowest parts of the mountain where species M used to grow become too warm for it, or other plants that grow better in warmer conditions move in and outcompete it. Species M now survives only higher up, where it is still cold enough.

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22 · Part 1 Paper 1 & 2 Human influences on the environment [7]

Mark scheme

(a) **[2]**

- oxygen concentration falls from 9.0 mg per dm^3 at the pipe to a minimum of 1.5 mg per dm^3 at 300 m downstream (1)
- then rises again to 8.0 mg per dm^3 at 1 000 m downstream (1)

(b) **[3]**

- sewage provides food (organic matter) for bacteria OR decomposers (1)
- so the bacteria multiply (1)
- the bacteria respire aerobically and use up the dissolved oxygen (1)

(c) **[1]**

- there is not enough oxygen for the mayfly larvae to respire (aerobically), so they die OR move away (1)

(d) **[1]**

- sewage contains pathogens OR disease-causing micro-organisms (1)

ALLOW named examples such as bacteria that cause disease

Worked solution

(a) At the pipe the water has 9.0 mg per dm^3 of oxygen. This falls to 3.0 at 100 m and to the lowest value, 1.5 mg per dm^3, at 300 m. Further downstream it recovers to 4.5 at 600 m and 8.0 mg per dm^3 at 1 000 m.

(b) Sewage contains organic matter, which is food for bacteria. The bacteria grow and divide rapidly. They respire aerobically and take dissolved oxygen from the water faster than it is replaced, so the oxygen concentration falls.

(c) Mayfly larvae need dissolved oxygen for aerobic respiration. At 100 m and 300 m the oxygen concentration is very low, so the larvae die or move away.

(d) Sewage can contain pathogens (disease-causing bacteria and viruses), so people who drink or swim in the water may become ill.

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23 · Part 1

Paper 1 & 2 Human influences on the environment [8]

Mark scheme

(a) **[6]**

Any six of:

- rain leaches nitrates (mineral ions) from the field into the lake (1)
- algae grow rapidly OR an algal bloom forms (1)
- the algae block light from reaching plants below the surface (1)
- so these plants cannot photosynthesise and die (1)
- algae also die (1)
- decomposers OR bacteria decompose the dead plants and algae and increase in number (1)
- the decomposers respire aerobically and use up the oxygen in the water (1)
- fish cannot get enough oxygen for respiration, so they die (1)

(b) **[2]**

Any two of:

- use less fertiliser (1)
- do not spread fertiliser when heavy rain is forecast (1)
- leave a strip of land with plants (e.g. hedges or grass) between the field and the lake (1)
- use manure OR slow-release fertiliser instead (1)

Worked solution

(a) Heavy rain washes (leaches) nitrates from the fertiliser out of the soil and into the lake. The extra nitrates let algae grow very quickly, forming a thick layer at the surface. This blocks light, so the plants under the water cannot photosynthesise and they die. Many algae die too. Decomposers feed on all this dead material and increase in number. They respire aerobically and use up the dissolved oxygen in the lake. The fish cannot get enough oxygen for respiration, so they die. This sequence is called eutrophication.

(b) The farmer could use less fertiliser, avoid spreading it just before heavy rain, or leave a strip of plants between the field and the lake to take up nitrates before they reach the water.

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24 · Part 1

Paper 2 only

Human influences on the environment [6]

Mark scheme

(a) **[3]**

- heavy rain leaches (washes) mineral ions out of the soil (1)
- there are no tree roots to absorb the mineral ions before they are washed away (1)
- fewer dead leaves fall, so decomposers return fewer mineral ions to the soil (1)

ALLOW the crops are harvested and removed, so the mineral ions in them are not returned to the soil

(b) **[3]**

- fewer trees, so less transpiration OR less evapotranspiration (1)
- so less water vapour enters the atmosphere (1)
- so fewer clouds form and there is less rainfall (1)

Worked solution

(a) In a forest, tree roots take up mineral ions, and dead leaves are decomposed so their minerals return to the soil. When the trees are cut down, there are no roots to take up the minerals and few dead leaves to recycle them. Heavy rain washes (leaches) the mineral ions deep into the ground or into rivers. Harvesting crops also removes minerals. The soil becomes less fertile.

(b) Trees release large amounts of water vapour into the air by transpiration; together with evaporation from the ground this is evapotranspiration. With fewer trees, less water vapour enters the atmosphere, so fewer clouds form and less rain falls in the area.

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25 · Part 1 Paper 2 only Human influences on the environment [8]

Mark scheme

(a) **[2]**

- 2 400 × 500 (1)
- 1.2×10^6 (tonnes) (1)

ALLOW 1 200 000 for 1 mark only (not in standard form)

(b) **[4]**

- combustion of the trees releases carbon dioxide (1)
- the carbon was stored in the wood for many years (1)
- fewer trees, so less photosynthesis OR less carbon dioxide is removed from the atmosphere (1)
- decomposers break down the remaining dead wood OR roots and release carbon dioxide in respiration (1)

(c) **[2]**

- less oxygen is released into the atmosphere (1)
- because there is less photosynthesis (1)

Worked solution

(a) Mass not removed = 2 400 km^2 × 500 tonnes per km^2 = 1 200 000 tonnes = 1.2×10^6 tonnes. Check: $1.2 \times 10^6 \div 2\,400 = 500$ tonnes per km^2.

(b) Trees store carbon in their wood for many years. Burning them (combustion) releases this carbon quickly as carbon dioxide. With the trees gone, less photosynthesis takes place, so less carbon dioxide is taken out of the atmosphere. Roots and leftover wood are broken down by decomposers, which release more carbon dioxide in respiration. All of these increase the concentration of carbon dioxide in the atmosphere.

(c) Trees release oxygen in photosynthesis. Fewer trees means less photosynthesis, so less oxygen is released into the atmosphere and the balance of atmospheric gases changes.

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